Bankruptcy Prediction Pipeline

Polish Companies Bankruptcy Data · XGBoost with Optuna · Technical documentation

1. Project Overview

This project builds a machine learning pipeline that predicts corporate bankruptcy from financial ratios of Polish companies. The data cover 2000–2013 and come in five forecasting horizons. The final model uses the *3year* dataset, which predicts bankruptcy status three years ahead, and is designed as a first-stage screening filter for bank credit risk assessment: it should catch nearly all future bankruptcies and clear as many healthy companies as possible without manual review.

Dataset	Instances	Bankrupt	Healthy	Bankrupt share	Horizon
1year	7,027	271	6,756	3.9%	5 years
2year	10,173	400	9,773	3.9%	4 years
3year	10,503	495	10,008	4.7%	3 years
4year	9,792	515	9,277	5.3%	2 years
5year	5,910	410	5,500	6.9%	1 year

Table 1. Dataset summary. Each file contains 64 financial ratios (Attr1–Attr64) from one year of the forecasting period; the label is bankruptcy status after the stated horizon (UCI dataset description).

2. Evaluation Protocol and Data Leakage Fix

Each dataset is split into a stratified 80/20 train/test split (random_state=42) **before** any preprocessing. Every step that learns from data — winsorization bounds, the imputer, hyperparameters, the decision threshold and the model — is fitted on the training set only. The test set is used exactly once, for the final evaluation.

An earlier version of the pipeline violated this protocol in two ways: winsorization quantiles and the MICE imputer were fitted on the full dataset before splitting (preprocessing leakage), and the decision threshold of 0.1 was chosen by inspecting test-set metrics (test-set reuse). Both were corrected. With the same tuning setup, test ROC-AUC on the 3year dataset changed from 0.862 (leaky) to 0.852 (clean), indicating a small but real optimistic bias. Repeated Optuna runs vary test ROC-AUC by roughly ±0.01; all results below come from the final hyperparameters stored in best_params.json.

3. Preprocessing

3.1 Winsorization

Financial ratios have extreme tails caused by near-zero denominators (e.g. Attr37 reaches 398,920 against a median of about 1.5). Each feature is clipped at its 1st and 99th percentiles, computed on the training set; the same bounds are applied to the test set.

3.2 Missing values

Two features have substantial missingness: Attr37 (39%) and Attr21 (23%); all others are below 5%. Missing values are imputed with MICE (IterativeImputer, 5 iterations) using a Random Forest regressor as the estimator, fitted on the training set and applied to the test set. The strategy was chosen in a separate validation experiment that masked known values and measured reconstruction quality:

Strategy	Validation R^2	Decision
Median imputation	−0.096	Rejected
MICE + BayesianRidge	−51.94	Rejected
MICE + RandomForest	0.871	Selected

Table 2. Imputation strategy comparison (R^2 on masked observed values).

4. Model and Hyperparameter Optimization

The classifier is XGBoost. Hyperparameters are tuned separately for each horizon with Optuna (TPE sampler, 50 trials, 5-fold stratified cross-validation on the training set). The objective is the F2 score, which weights recall twice as heavily as precision, reflecting that a missed bankruptcy costs a lender far more than an unnecessary manual review. Class imbalance is handled through scale_pos_weight, tuned as a hyperparameter, rather than by resampling.

Parameter	Search range	Role
n_estimators	100 – 1000	Number of trees
max_depth	3 – 10	Tree depth
learning_rate	0.01 – 0.3 (log scale)	Shrinkage per boosting step
subsample	0.5 – 1.0	Row sampling per tree
colsample_bytree	0.5 – 1.0	Feature sampling per tree
min_child_weight	1 – 10	Minimum child weight
gamma	0 – 5	Minimum loss reduction to split
scale_pos_weight	1× – 3× class ratio	Class imbalance correction

Table 3. Optuna search space.

5. Decision Threshold Selection

The operating threshold is selected without using the test set. Five-fold out-of-fold (OOF) predicted probabilities are generated on the training set, so that every training company receives a prediction from a model that did not see it. The business requirement for a screening filter is to catch at least 90% of bankruptcies; the threshold is therefore set to the **highest value at which OOF recall is at least 90%**, which maximizes the number of companies cleared automatically subject to that constraint. The selected threshold is **0.04** (OOF recall 92.2%).

Threshold	Precision	Recall	F2	Flagged
0.05	0.096	0.899	0.336	44.2%
0.10	0.121	0.813	0.378	31.8%
0.20	0.163	0.705	0.423	20.4%
0.30	0.204	0.593	0.430	13.7%
0.50	0.307	0.465	0.421	7.1%

Table 4. Out-of-fold precision–recall trade-off on the 3year training set. Raising the threshold reduces analyst workload at the cost of missed bankruptcies.

6. Results

6.1 Final model (3year dataset, test set, threshold 0.04)

	Predicted: not bankrupt	Predicted: bankrupt
Actual: not bankrupt	1,002 (TN)	1,000 (FP)
Actual: bankrupt	6 (FN)	93 (TP)

Table 5. Confusion matrix on the test set (2,101 companies, 99 bankrupt).

Metric	Value
ROC-AUC	0.865
PR-AUC (average precision)	0.390
Recall	93.9% (93 of 99)
Precision	8.5%
F2	0.312
Companies flagged for review	1,093 of 2,101 (52.0%)
Companies cleared automatically	1,008 of 2,101 (48.0%)
Bankruptcies missed	6 of 99 (6.1%)

Table 6. Screening summary. Test recall (93.9%) is consistent with the OOF estimate used to select the threshold (92.2%).

6.2 Performance across horizons

Dataset	Horizon	ROC-AUC	PR-AUC	Bankrupt share
1year	5 years	0.862	0.275	3.9%
2year	4 years	0.830	0.300	3.9%
3year	3 years	0.865	0.390	4.7%
4year	2 years	0.830	0.359	5.3%
5year	1 year	0.901	0.486	6.9%

Table 7. Threshold-free test metrics per horizon. The baseline PR-AUC of a random classifier equals the bankrupt share, so PR-AUC is only roughly comparable across rows.

Discrimination is strongest one year before bankruptcy, when financial distress is most visible in the ratios. The 3year horizon is used for the final model as a compromise between early warning and predictive power. Threshold-dependent metrics are not compared across horizons, because a single fixed threshold corresponds to very different operating points for models with different probability scales.

6.3 Comparison with Random Forest baseline

Metric	XGBoost	Random Forest
ROC-AUC	0.865	0.828
PR-AUC	0.390	0.235
Recall @ 0.04	0.939	0.899
Precision @ 0.04	0.085	0.101

Table 8. XGBoost versus a Random Forest baseline (100 trees, class-weighted) on the 3year test set.

XGBoost is the stronger discriminator on both threshold-free metrics; its PR-AUC, the most informative metric under a 5% positive rate, is about 1.7 times that of the baseline. At a shared threshold the Random Forest shows slightly higher precision, but point metrics at a common threshold are not a fair comparison

because the two models produce differently calibrated probabilities. XGBoost is therefore selected as the final model.

7. Feature Importance (SHAP)

Feature contributions are computed with SHAP (TreeExplainer) on the test set. The three dominant drivers are:

Rank	Feature	Financial meaning
1	3-year gross profit / total assets (Attr24)	Long-term profitability
2	(Net profit + depreciation) / total liabilities (Attr26)	Debt repayment capacity
3	(Current assets – inventory) / short-term liabilities (Attr46)	Short-term liquidity

Table 9. Top SHAP features (mean absolute SHAP value).

Further down the ranking appear retained earnings, asset turnover, operating profit relative to financial expenses, sales growth and firm size (log of total assets). The results are economically consistent: profitability, liquidity and debt coverage — the factors a credit analyst assesses first — drive the predictions.

8. Limitations

– **Low precision by design.** About 1 in 12 flagged companies goes bankrupt. This is acceptable for a first-stage filter followed by manual review, not for automatic rejection.

– **Random split.** Train and test companies come from the same period; a time-based split would better reflect performance on future years.

– **Historical data.** Polish companies from 2000–2013; default patterns may differ today.

– **Preprocessing inside cross-validation.** The imputer is fitted on the full training set, so the CV folds used for tuning and threshold selection share it. This does not affect the test evaluation; wrapping preprocessing in a scikit-learn Pipeline would remove it.

– **Tuning variance.** Optuna results vary between runs (about ±0.01 ROC-AUC); the search is now seeded for reproducibility.